INCH-POUND
MIL-PRF-1/894G
17 January 2014
SUPERSEDING
MIL-PRF-1/894F
10 April 2007

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 884

INACTIVE FOR NEW DESIGN
AFTER 30 APRIL 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification and MIL-PRF-1.

DESCRIPTION: Triode.

Mounting position: Any.

Weight: 1.5 ounces (42.5 grams) nominal.

Envelope: ST-12 (EIA).

Base: B6-3 (EIA).

Height: 4.125 inches (104.78 mm) maximum.

Diameter: 1.563 inches (39.70 mm) maximum.

Pin connections:

Pin no.:	1	2	3	5	7	8
Element:	nc	h	a	g	h	k

ABSOLUTE RATINGS:

Parameter:	Ef	epx or epy	Peak voltage between any two electrodes	ib	Ib	Rg	tk	Ehk	TA
Unit:	V	v	v	mA	mA dc (Note 1)	MegΩ (Note 2)	s	V dc (Note 3)	°C
Oscillator:									
Maximum:	6.93	300	350	300	2	0.5	30	-100	+85
Minimum:	5.67	---	---	---	---	---	---	---	-60
Rectifier:									
Maximum:	6.93	300	350	300	75	0.5	30	-100	+85
Minimum:	5.67	---	---	---	---	---	---	---	-60
Test conditions:	6.3	---	---		---	---	30	---	---

GENERAL:

Qualification - Not required.

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

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TABLE I. Requirements or tests.

Requirement or test	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 1							
Heater current	3241			If	540	660	mA
Critical grid voltage for conduction (1)	3201	4/	Eb = 110 V dc; Rp = 1,700 ohms; Rg = 0.1 MegΩ	Ec(1)	-9	-16	V dc
Critical grid voltage for conduction (2)	3201		Eb = 250 V dc; Rp = 5,600 ohms; Rg = 0.1 MegΩ	Ec(2)	-21	-31	V dc

Pulse emission	1231		epp = eggl = 180 ± 9.0 v;
			prr = 100 ± 5;
			tk = 5 (min);
			Ra = 10 ohms;
			Rp = 20 ohms;
			calibrating resistor = 10 ohms;
			Zm = 7.5;
			t = 3 (max)
				etd	---	69	v

Conformance inspection, part 2							
Low frequency vibration	1031		No voltages applied	---	---	---	---
Heater-cathode leakage	1336			Ihk	0	20	μA dc
Conformance inspection, part 3							
Life test	---		Group B (see figure 1); t = 500 hours	---	---	---	---
Life-test end points:	---						
Critical grid voltage for conduction (1)	3201		Eb = 110 Vdc; Rp = 1,700 ohms; Rg = 0.1 MegΩ	Ec(1)	-7.0	-18.0	V dc
Critical grid voltage for conduction (2)	3201		Eb = 250 Vdc; Rp = 5,600 ohms; Rg = 0.1 MegΩ	Ec(2)	-19.0	-33.0	V dc

Pulse emission	1231		epp = eggl = 180 ± 9.0 v;
			prr = 100 ± 5;
			tk = 5 (min);
			Ra = 10 ohms;
			Rp = 20 ohms;
			calibrating resistor = 10 ohms;
			Zm = 7.5;
			t = 3 (max)
				etd	---	110	v

See footnotes at top of next page.

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TABLE I. Requirements or tests - Continued.

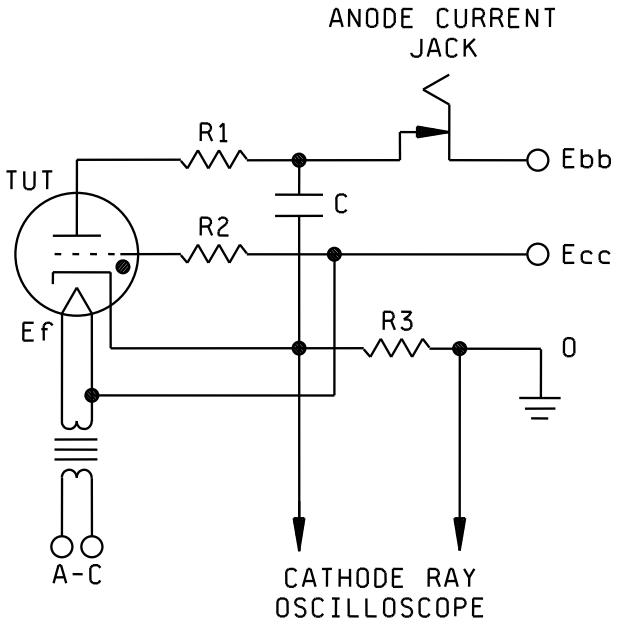
NOTES:

1/ Below 200 Hz, Ib = 3 mA dc (maximum).

2/ Grid resistance shall be not less than 1,000 ohms per volt (maximum instantaneous) applied to grid.

3/ Peak voltage between any electrode and heater shall not exceed 350 volts.

4/ This test shall be the first test performed at the conclusion of the holding period.



Ebb = 300 V dc
Ecc = 150 V dc
R1 = 500 OHMS
R2 = 30,000 OHMS
R3 = 330,000 OHMS
C = 0.005 μF

PEAK VOLTAGE - 136 VOLTS
PEAK CURRENT - 250 mA
FREQUENCY - 1,000 Hz
AVERAGE CURRENT - 0.7 mA dc

FIGURE 1. Relaxation oscillator test circuit.

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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5960-2013-044)

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